Hochschule Darmstadt

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Computer Science Department

Introduction to AI

Lab 4 Report

Author:

Georgios Karyos

Pantelis Stylianou

Lecturer:

Dr. Hamed Shariat Yazdi

a)

Example Queries

Selection of triples | Selection of classes

Prefixes

rdf | rdfs | owl | xsd

SPARQL Endpoint: /TMDB/query

Content Type (SELECT): JSON

Content Type (GRAPH): Turtle

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT DISTINCT ?name
4  WHERE {
5      ?g a :Genre ;
6         :name ?name .
7  }
```

name
1Comedy
2Drama
3Family
4Romance
5Thriller
6Action
7Animation
8Adventure
9Horror
10Documentary
11Music
12Crime
13Science Fiction
14Mystery
15Foreign
16Fantasy
17War
18Western
19History
20TV Movie

b)

Example Queries

Selection of triples | Selection of classes

Prefixes

rdf | rdfs | owl | xsd

SPARQL Endpoint: /TMDB/query

Content Type (SELECT): JSON

Content Type (GRAPH): Turtle

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT DISTINCT ?dept
4  WHERE {
5      ?c a :Crew ;
6         :department ?dept .
7  }
```

dept
1Directing
2Writing
3Production
4Sound
5Camera
6Editing
7Art
8Costume & Make-Up
9Crew
10Lighting
11Visual Effects
12Actors

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c)

Example Queries

Selection of triples | Selection of classes

Prefixes

rdf | rdfs | owl | xsd

SPARQL Endpoint: /TMDB/query

Content Type (SELECT): JSON

Content Type (GRAPH): Turtle

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT (COUNT(?comp) AS ?numCompanies)
4  WHERE {
5      ?comp a :Company .
6  }
7
```

Table | Response | 1 result in 0.28 seconds

Simple view ☐ Ellipse ☑ Filter query results Page size: 50

numCompanies
1"3712"^^<http://www.w3.org/2001/XMLSchema#integer>

Showing 1 to 1 of 1 entry << < 1 > >>

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d)

Example Queries

Selection of triples | Selection of classes

Prefixes

rdf | rdfs | owl | xsd

SPARQL Endpoint: /TMDB/query

Content Type (SELECT): JSON

Content Type (GRAPH): Turtle

```
PREFIX : <https://www.themoviedb.org/kaggle-export/>

SELECT ?id ?name
WHERE {
    ?c a :Collection ;
       :id ?id ;
       :name ?name .
```

id	name
1"313576"^^<http://www.w3.org/2001/XMLSchema#integer>	Hot Tub Time Machine Collection
2"107674"^^<http://www.w3.org/2001/XMLSchema#integer>	The Princess Diaries Collection
3"256377"^^<http://www.w3.org/2001/XMLSchema#integer>	The Muppet Collection
4"1575"^^<http://www.w3.org/2001/XMLSchema#integer>	Rocky Collection
5"48190"^^<http://www.w3.org/2001/XMLSchema#integer>	Revenge of the Nerds Collection
6"91698"^^<http://www.w3.org/2001/XMLSchema#integer>	Chili Palmer Collection
7"9518"^^<http://www.w3.org/2001/XMLSchema#integer>	The Transporter Collection
8"9735"^^<http://www.w3.org/2001/XMLSchema#integer>	Friday the 13th Collection
9"207621"^^<http://www.w3.org/2001/XMLSchema#integer>	V/H/S Collection
10"207632"^^<http://www.w3.org/2001/XMLSchema#integer>	The ABCs of Death Collection
11"90863"^^<http://www.w3.org/2001/XMLSchema#integer>	Rush Hour Collection
12"376970"^^<http://www.w3.org/2001/XMLSchema#integer>	Cocaine Cowboys
13"8354"^^<http://www.w3.org/2001/XMLSchema#integer>	Ice Age Collection
14"115570"^^<http://www.w3.org/2001/XMLSchema#integer>	Star Trek: The Next Generation Collection
15"120794"^^<http://www.w3.org/2001/XMLSchema#integer>	Batman Collection
16"645"^^<http://www.w3.org/2001/XMLSchema#integer>	James Bond Collection
17"36694"^^<http://www.w3.org/2001/XMLSchema#integer>	Bloodsport Collection
18"9485"^^<http://www.w3.org/2001/XMLSchema#integer>	The Fast and the Furious Collection

e)

SPARQL Query

To try out some SPARQL queries against the selected dataset, enter your query h

Example Queries

Prefi

Selection of triples **Selection of classes**

rdf

SPARQL Endpoint

/TMDB/query

Content Type (SELECT)

JSON

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT ?German_movie ?title
4  WHERE {
5      ?German_movie a :Movie ;
6              :original_language "de" ;
7              :title ?title .
8  }
9
```

German_movie	title
1<https://www.themoviedb.org/kaggle-export/Movie_248>	Downfall
2<https://www.themoviedb.org/kaggle-export/Movie_727>	Barbara
3<https://www.themoviedb.org/kaggle-export/Movie_763>	The Lives of Others
4<https://www.themoviedb.org/kaggle-export/Movie_872>	Rabbit Without Ears 2
5<https://www.themoviedb.org/kaggle-export/Movie_961>	Fuck You Goethe 2
6<https://www.themoviedb.org/kaggle-export/Movie_1105>	Robinson Crusoe
7<https://www.themoviedb.org/kaggle-export/Movie_1132>	Toni Erdmann
8<https://www.themoviedb.org/kaggle-export/Movie_1186>	Faraway, So Close!
9<https://www.themoviedb.org/kaggle-export/Movie_1277>	The Red Baron
10<https://www.themoviedb.org/kaggle-export/Movie_1380>	The Edge of Heaven
11<https://www.themoviedb.org/kaggle-export/Movie_2026>	Look Who's Back
12<https://www.themoviedb.org/kaggle-export/Movie_2272>	Before the Fall
13<https://www.themoviedb.org/kaggle-export/Movie_2303>	Happy Weekend
14<https://www.themoviedb.org/kaggle-export/Movie_2366>	Labyrinth of Lies
15<https://www.themoviedb.org/kaggle-export/Movie_2496>	Wings of Desire
16<https://www.themoviedb.org/kaggle-export/Movie_2509>	Pina
17<https://www.themoviedb.org/kaggle-export/Movie_2865>	The Goalie's Anxiety at the Penalty Kick
18<https://www.themoviedb.org/kaggle-export/Movie_2886>	13 Minutes

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f)

Selection of triples | Selection of classes | rdf | rdfs | owl

SPARQL Endpoint

/TMDB/query

Content Type (SELECT)

JSON

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT DISTINCT ?movie ?title
4  WHERE {
5      ?movie a :Movie ;
6             :title ?title ;
7             :cast ?c .
8      ?c :character ?char .
9      FILTER CONTAINS(LCASE(?char), "julius caesar")
10 }
11
```

Table | Response | 2 results in 2.122 seconds

Simple view ☐ Ellipse ☑ Filter query results Page size: 50

	movie	title
1	<https://www.themoviedb.org/kaggle-export/Movie_1923>	Cleopatra
2	<https://www.themoviedb.org/kaggle-export/Movie_2121>	Spartacus

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g)

Selection of triples | Selection of classes | rdf | rdfs | owl | xsd

SPARQL Endpoint: /TMDB/query

Content Type (SELECT): JSON

Content Type (GRAPH): Turtle

```
PREFIX : <https://www.themoviedb.org/kaggle-export/>

SELECT DISTINCT ?name
WHERE {
    ?cast a :Cast ;
          :id ?pid ;
          :name ?name .

    ?crew a :Crew ;
          :id ?pid ;
          :job "Director" .
}
```

name
1Garry Marshall
2Brian Henson
3Christopher Guest
4F. Gary Gray
5Steven Spielberg
6Wes Craven
7Ralph Fiennes
8Joe Swanberg
9Matt Bettinelli-Olpin
10Ti West
11Adam Wingard
12Tyler Gillett
13Chad Villella
14Larry Fessenden
15Julian Barratt
16Brett Ratner
17Carlos Saldanha
18Quentin Tarantino

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h)

Selection of triples | Selection of classes | rdf | rdfs

SPARQL Endpoint

/TMDB/query

Content Type (SELECT)

JSON

```
PREFIX : <https://www.themoviedb.org/kaggle-export/>

SELECT ?movieTitle ?character
WHERE {
    ?cast a :Cast ;
          :name "Leonardo DiCaprio" ;
          :character ?character .

    ?movie :cast ?cast ;
           :title ?movieTitle .
}

```

movieTitle	character
1The Quick and the Dead	Fee Herod 'The Kid,Äú
2Gangs of New York	Amsterdam Vallon
3Don's Plum	
4The Revenant	Hugh Glass
5Blood Diamond	Danny Archer
6This Boy's Life	Tobias Wolff
7Marvin's Room	Hank
8The Aviator	Howard Hughes
9What's Eating Gilbert Grape	Arnie Grape

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i)

SPARQL Endpoint

/TMDB/query

Content Type (SELECT)

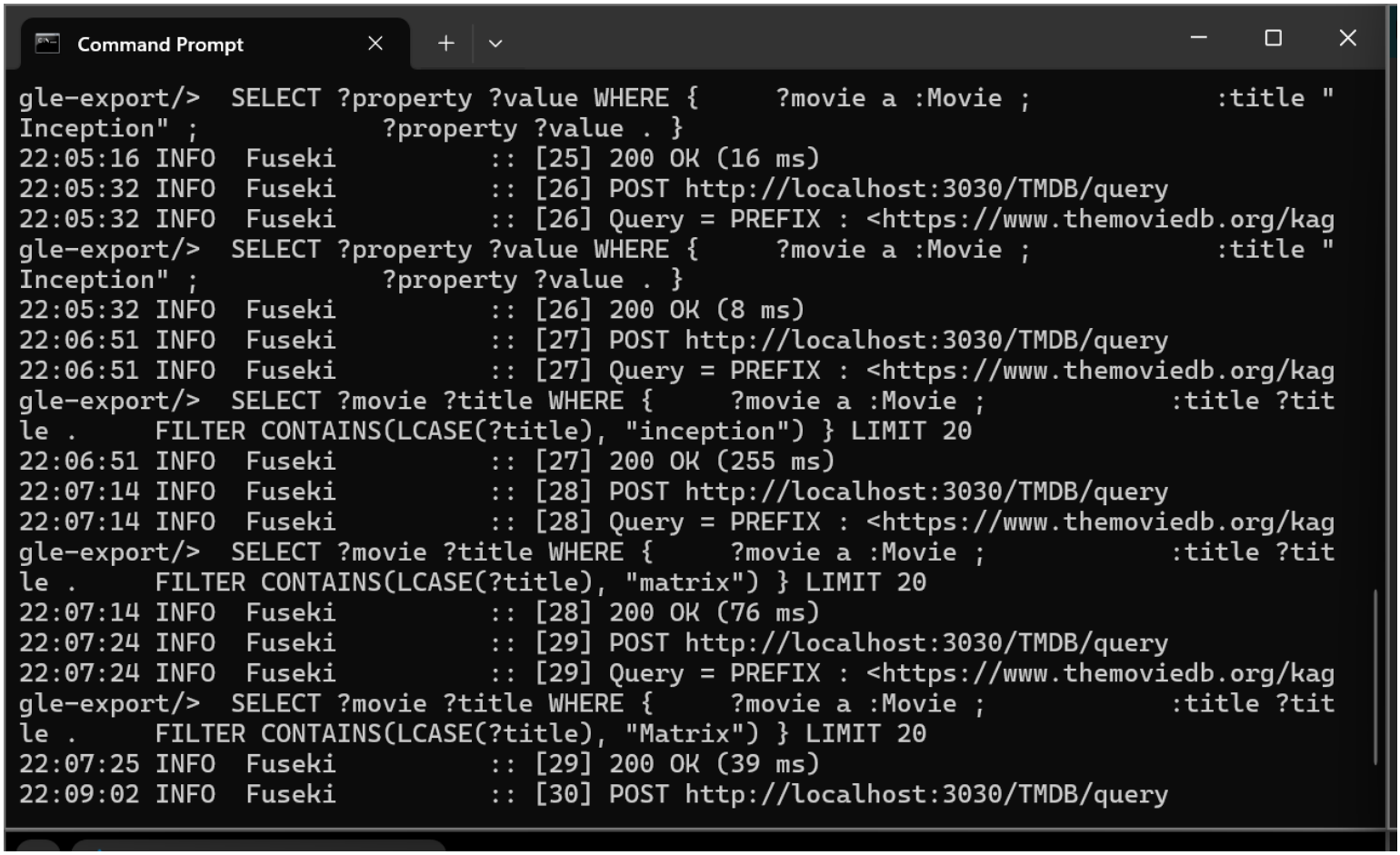
JSON

```
1  PREFIX : <https://www.themoviedb.org/kaggle-export/>
2
3  SELECT ?property ?value
4  WHERE {
5      ?m a :Movie ;
6         :title ?t ;
7         ?property ?value .
8      FILTER(STR(?t) = "Downfall")
9  }
```

	property	value
1	<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>	<https://www.themoviedb.org/kaggle-export/Movie>
2	<https://www.themoviedb.org/kaggle-export/id>	"248"^^<http://www.w3.org/2001/XMLSchema#integer>
3	<https://www.themoviedb.org/kaggle-export/poster_pa...	/2QmpblYvUThzBQ4ArdT1Kp43DZi.jpg
4	<https://www.themoviedb.org/kaggle-export/budget>	"18339750"^^<http://www.w3.org/2001/XMLSchema#integer>
5	<https://www.themoviedb.org/kaggle-export/genres>	<https://www.themoviedb.org/kaggle-export/Genre_18>
6	<https://www.themoviedb.org/kaggle-export/genres>	<https://www.themoviedb.org/kaggle-export/Genre_10...
7	<https://www.themoviedb.org/kaggle-export/genres>	<https://www.themoviedb.org/kaggle-export/Genre_36>
8	<https://www.themoviedb.org/kaggle-export/imdb_id>	tt0363163
9	<https://www.themoviedb.org/kaggle-export/original_la...	de
10	<https://www.themoviedb.org/kaggle-export/original_ti...	Der Untergang
11	<https://www.themoviedb.org/kaggle-export/overview>	In April of 1945, Germany stands at the brink of defeat w...
12	<https://www.themoviedb.org/kaggle-export/popularity>	"7.69224"^^<http://www.w3.org/2001/XMLSchema#float>
13	<https://www.themoviedb.org/kaggle-export/productio...	<https://www.themoviedb.org/kaggle-export/Company...
14	<https://www.themoviedb.org/kaggle-export/productio...	<https://www.themoviedb.org/kaggle-export/Company...
15	<https://www.themoviedb.org/kaggle-export/productio...	<https://www.themoviedb.org/kaggle-export/Company...

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Terminal Server Log:



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